

System Interface

Business interface, 'IComparisonService'

This is the primary interface used by the Presentation layer(controllers). It encapsulate the core functional requirements of the application.

The 'getSharePrices(String symbol, LocalDate start, LocalDate end) : List<SharePrice.>', purpose is to retrieve the daily price data for a single symbol. So it enforces the two-year maximum date range validation. It coordinates between the cache(IPriceRepository) and the live provider(IExternalPriceProvider).

The 'compareShares(String symbol1, String symbol2, LocalDate start, LocalDate end) : ComparisonResult', purpose is to retrieve and aligns data for two companies to allow for graphical comparison.

Persistence interface, 'IStockRepository'

The persistence interface manages the "Persistence store" requirement.

The 'findCachedPrices(String symbol, LocalDate start, LocalDate end) : List<SharePrice.>', checks the local persistent store like JSON or Database for existing data to enable offline functionality.

The 'savePrices(List<SharePrice.> prices) : void', persists newly fetched data from the external API into the local store.

Data Source Interface, 'IPriceData'

This interface abstracts the data source such as Yahoo finance.

The, 'fetchRemotePrices(String symbol, LocalDate start, LocalDate end) : List<SharePrice.>', performs the network call to obtain raw historical data from the provider.

Validation Interface, 'IValidator'

The, 'validateRequest(symbol, start, end) : boolean', specifically enforces the two-year maximum range and verifies ticker symbols formats before any fetching occurs.

Allocation to components

Component	Implements Interface	Description
ShareService Component	IComparisonService	Acts as the "Interactor". Contains the logic for the 2-year range limit and handles the logic of "Cache-Aside" (check repo, then fetch remote).
Persistence Component	IStockRepository	Responsible for I/O operations. Can be swapped between a JsonPriceRepository and a SQLitePriceRepository without changing business logic.
Network Component	IPriceData	Uses the HttpClient to communicate with the external financial API. Converts raw JSON responses into internal SharePrice objects.
Validation Service	IValidator	Responsible for ensuring all user inputs meet business constraints. It strictly enforces the mandatory two-year limit on date ranges and verifies ticker symbol formats before any data retrieval or comparison processes begin.

Collaboration Diagram

